ModelIt!

Student Activity Pack

Climate Tipping Points: The Point of No Return

When Earth's Climate System Crosses a Threshold It Can't Come Back From

Grade 09L2 | NGSS HS-ESS3-5, HS-ESS2-4 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you already know about climate change? What causes it?

2 What does 'tipping point' mean to you? Can you think of an everyday example where a small change triggers a big, irreversible effect?

3 Why do you think scientists talk about 1.5°C as a critical number for global warming?

4 Draw what you think happens to Earth's systems if average temperatures increase by 2°C.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

CO2 Emissions | Global Temperature | Ice Cover | Albedo Effect | Permafrost Methane Release | Sea Level | Ocean Heat Absorption

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Below Threshold

Moderate CO_2 emissions, current conditions

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Crossing the Threshold

CO_2 emissions increase past the tipping point

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Attempted Recovery

Emissions drop to zero after tipping point is crossed

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Carbon Sink Capacity — The ability of forests and oceans to absorb CO_2, which decreases as these systems are stressed or destroyed

Thermohaline Circulation — The global ocean current system driven by temperature and salinity differences, which could collapse if freshwater from melting ice disrupts the pattern

Cloud Feedback — Changes in cloud formation patterns that can either amplify or dampen warming, one of the biggest uncertainties in climate models

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Is there a temperature where Earth's climate spirals out of control — and are we close to it? Use evidence from your simulation to support your answers.

1. **What makes a tipping point different from a gradual change? How does your model demonstrate this difference?**

2. **Why is thermal inertia (time delay) one of the most dangerous features of the climate system?**

3. **Your model shows that reducing emissions to zero after crossing the tipping point doesn't restore the system. Why not?**

4. **How do the ice-albedo and permafrost methane feedback loops reinforce each other? What does cascading tipping points mean?**

5. **If you were advising world leaders, what would you tell them based on your model's predictions?**

8 STEM Challenge: Design a Climate Tipping Point Data Dashboard

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YOUR ENGINEERING MISSION

Create an interactive data dashboard that visualizes the key climate tipping points, shows where we currently stand relative to each threshold, and communicates the urgency and irreversibility of crossing these points.

THE SCENARIO

A city government is making decisions about climate policy and needs a clear, visual tool that shows elected officials — who are not scientists — exactly what climate tipping points are, how close we are to them, and what happens if we cross them. Your team has been hired to build this dashboard using your model data.

Engineering Design Thinking

1. HOW WILL YOU REPRESENT MULTIPLE FEEDBACK LOOPS VISUALLY WITHOUT OVERWHELMING THE VIEWER?
2. WHAT COLOR CODING OR VISUAL LANGUAGE WILL YOU USE TO COMMUNICATE URGENCY LEVELS?
3. HOW WILL YOU SHOW THE TIME DELAY BETWEEN EMISSIONS AND EFFECTS (THERMAL INERTIA)?
4. HOW WILL YOUR DASHBOARD HELP A NON-SCIENTIST UNDERSTAND WHY 1.5°C VS. 2.0°C MATTERS SO MUCH?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Climate Scientists and Earth System Modelers build computational models of global climate systems and study tipping points. They work at NOAA, NASA, IPCC, universities, and think tanks, earning $80,000–$160,000/year. Their work directly informs international climate policy decisions.